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Simulation-Based Handover Decision Evaluation for High-Mobility Users

Suraj M

surajmahesh2654@gmail.com

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Problem Description

Fixed handover rules struggle as user speed increases

- Handover must be well timed when users move fast between cells
- Too late: handover failure and a dropped connection
- Too early or repeated: ping-pong between cells and wasted signalling
- Fixed signal-based rules do not adapt to user speed
- Goal: test, reproducibly in simulation, whether a learned policy beats the signal-based baseline

In scope

- One LTE/5G-capable simulator
- 3 cells, 2 user speeds
- One supervised model vs a signal-based baseline
- Multi-seed evaluation

Out of scope

- 6G standards-conformance claims, a new radio stack, deep RL, field deployment

Background

What is needed to understand the problem and the proposed optimization

- Handover (HO): moving a user's connection to a new cell as they move
- RSRP: measured signal power, the basis of HO decisions
- Baseline rule: switch when a neighbour beats the serving cell by a margin for a set time-to-trigger [1], [2]
- Failure: HO happens too late and the link drops first
- Ping-pong: quickly bouncing back to the old cell
- Speed: faster users leave less time to decide

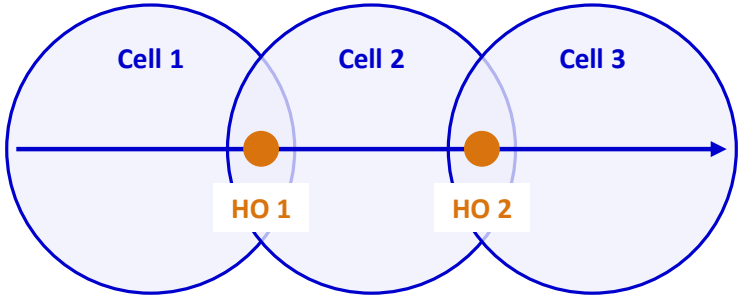


Figure: a user crossing three cells

Literature & RFC Review

The most relevant standards, tools and studies

Ref	Source	Relevance to this work
[1]	3GPP TS 36.331 (LTE RRC)	Defines measurement events such as A3 (neighbour better than serving) used by conventional handover
[2]	ns-3 LTE module: A3RsrpHandoverAlgorithm	Ready-made A3-RSRP rule with hysteresis and TTT; our baseline reference
[3]	Mollel et al.: survey of ML for handover management in 5G and beyond	Overview of ML-based handover work; context for our supervised approach
[4]	5G-LENA: 5G NR module for ns-3	Alternative NR simulator option; the radio model used must be stated clearly
[5]	RFC 2330: Framework for IP Performance Metrics	Guidance on precisely defined, repeatable metrics; applied to our evaluation design

Proposed Solution

Compare a learned handover policy with a signal-based baseline in one reproducible simulator

1 Simulate

ns-3 LTE module with 3 cells and 2 user speeds (e.g. slow and fast). Fixed seeds and saved configs.

2 Baseline

A3-RSRP rule with margin, hysteresis and TTT. Metrics logged for every run.

3 Learn

Log RSRP, trends and speed. Label the best target cell. Train one supervised model (decision tree or gradient boosting).

4 Evaluate

Run both policies over multiple seeds. Compare failures, ping-pong, interruption and throughput.

The trained model is exported into a custom ns-3 handover algorithm, so the baseline and the learned policy run in the same simulation.

Optimization Proposed

Proposed enhancements over the signal-based baseline

Aspect	Signal-based baseline (before)	Proposed learned policy (after)
Decision inputs	RSRP difference only	RSRP trend, signal quality, speed and neighbour history
Parameters	Fixed margin, hysteresis and TTT	Decision rule learned from simulated data
Speed handling	Same rule at every speed	Speed is an input, so decisions can adapt
Ping-pong control	Relies on hysteresis and TTT	Dwell-time features discourage returns to the previous cell
Evaluation	Single-scenario runs	Multi-seed mean and spread, reproducible

These are expected benefits to be tested. The experiments will show whether they hold.

Solution Architecture & Design

Components and how the optimization process flows between them

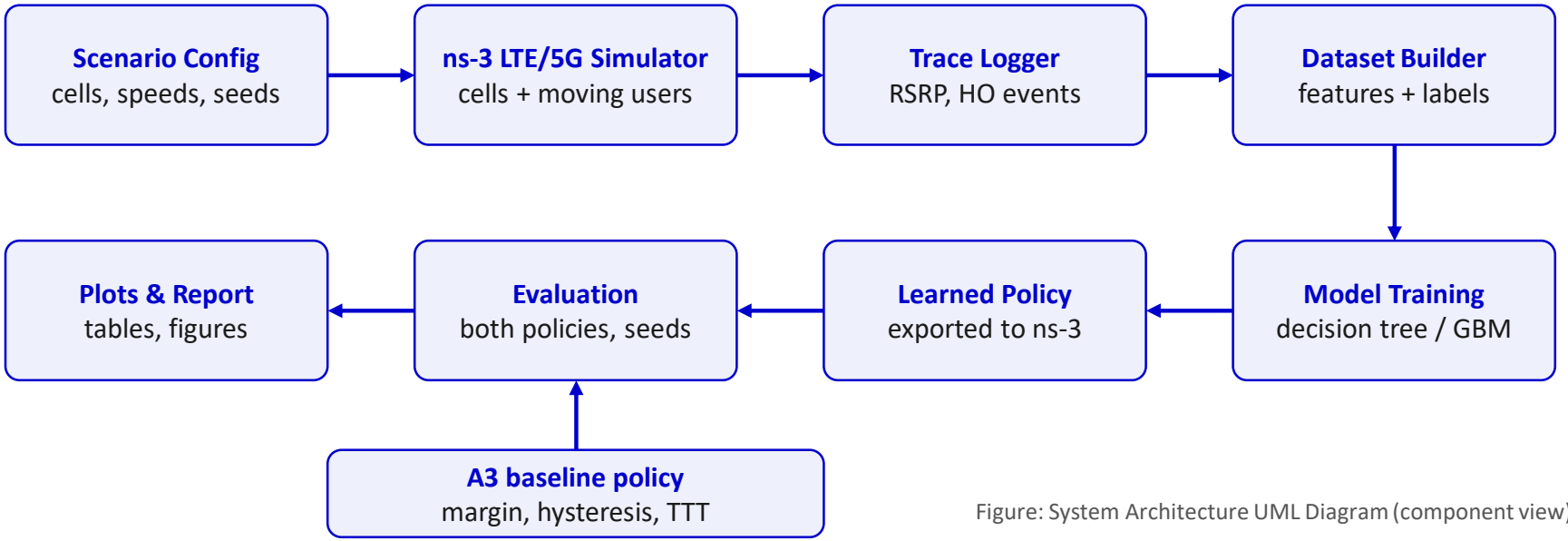


Figure: System Architecture UML Diagram (component view)

Data Flow Diagram

How data moves from simulation to results

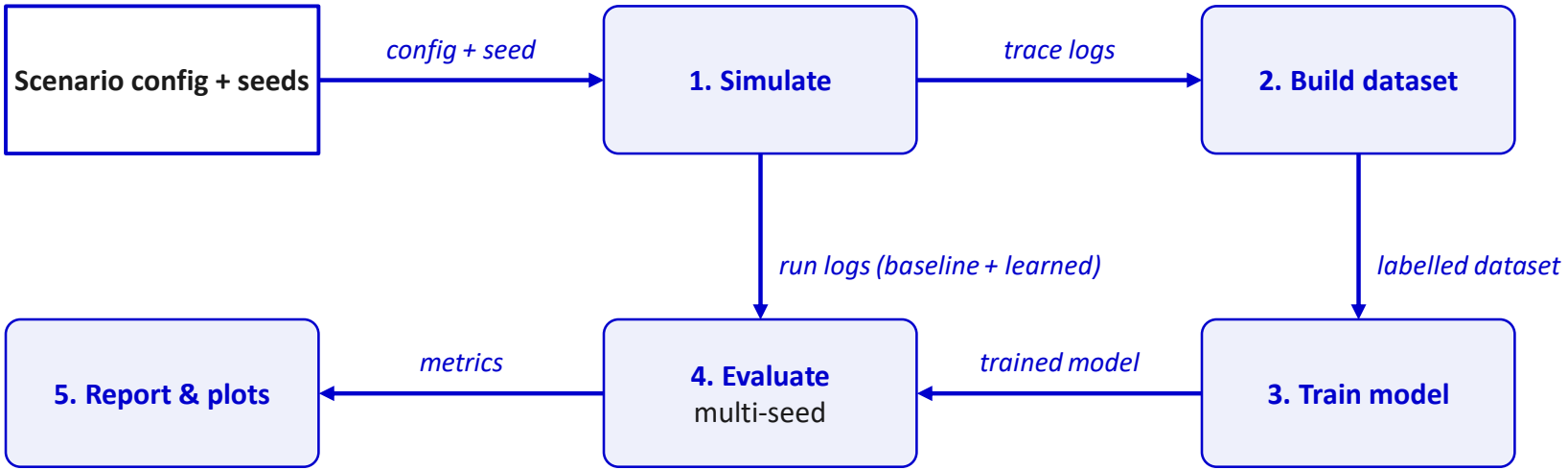


Figure: Data Flow Diagram (DFD)

Deployment Timeline

Six milestones over about three weeks, with mentor review at each one

1. **Week 1: Setup**
   Install the simulator, build the 3-cell scenario with 2 user speeds

2. **Weeks 1-2: Baseline**
   A3 handover rule, metric logging, first multi-seed runs

3. **Week 2: Dataset**
   Extract features, define labels, check data quality

4. **Weeks 2-3: Learned model**
   Train and tune one supervised model, export it into ns-3

5. **Week 3: Evaluation**
   Policy-in-the-loop runs across seeds, plots and analysis

6. **Week 3: Delivery**
   Report in the prescribed structure, repo clean-up, demo with pseudocode

Bibliography

[1] 3GPP. “Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC); Protocol specification.” TS 36.331.
[2] ns-3 Consortium. “A3RsrpHandoverAlgorithm class reference.” ns-3 LTE module. nsnam.org/doxygen/d3/d97/classns3_1_1_a3_rsrp_handover_algorithm.html
[3] M. S. Mollel et al. “A Survey of Machine Learning Applications to Handover Management in 5G and Beyond.” IEEE Access, 2021.
[4] CTTC. “5G-LENA: The 5G NR module for the ns-3 simulator.” 5g-lena.cttc.es
[5] V. Paxson, G. Almes, J. Mahdavi, M. Mathis. “Framework for IP Performance Metrics.” RFC 2330, May 1998.

Thank You for Your Attention!